



GLUCOSATIN

DRY POWDER FOR ORAL SOLUTION

COMPOSITION:

Each sachet contains:

Glucosamine Sulphate Potassium Chloride	1500mg
equivalent to Glucosamine Sulphate	
Chondroitin Sulphate Sodium	1200mg

PRESENTATION:

White to off-white granules, with orange spot colour, which gives an orange solution with orange flavour when reconstituted.

INDICATION:

As an adjunctive therapy for osteoarthritis.

PHARMACOLOGY:

Mechanism of Action:

Glucosamine is a natural substance found in chitin, mucoproteins, and mucopolysaccharides. It is involved in the manufacture of glycosaminoglycan, which forms cartilage tissue in the body, glucosamine is also present in tendons and ligaments. Glucosamine must be synthesized by the body but the ability to do this decline with age. Glucosamine and its salts have therefore been advocated in the treatment of rheumatic disorders including osteoarthritis. Glucosamine also acts to improve the viscosity of synovial fluid by increasing synovial fluid production, thereby providing lubricant activity. Chondroitin sulphate, as a vital component of aggrecan, helps to maintain the structural integrity of cartilage tissue. Loss of chondroitin sulphate from joint cartilage leads to loss of cartilage resilience and shock-absorbing properties. Orally administered chondroitin sulphate leads to increase in hyaluronic acid production and viscosity of synovial fluid. Chondroitin sulphate also inhibits the enzymes involved in degradation of cartilage. Combination therapy with glucosamine and chondroitin helps to protect joint cartilage from damage (chondroprotection), reduces joint pain and improves joint function and mobility. Glucosamine and chondroitin combination is effective in all forms of degenerative joint diseases such as osteoarthritis and arthrosis, both acute and chronic, bringing relief or reduction of joint pains, improvement in joint function, and inhibition or reversal of the degenerative disease process.

Pharmacokinetics:

Following oral administration, glucosamine sulphate is well absorbed from the gastrointestinal tract. The absolute bioavailability is 26% due to first-pass effect in the liver. The gastrointestinal absorption is close to 90%. Glucosamine is not protein bound, but rather incorporates into plasma protein (primarily

globulins). The volume of distribution is 2.5 liters. Plasma levels of glucosamine reach a peak after 8-10 hours and then decline, exhibiting a half-life of 68 hours. The first-pass effect in the liver in which more than 70% of glucosamine is metabolized. About 10% is excreted in the urine, with about 11% found in the faeces. Part of dose of glucosamine sulphate is eliminated as carbon dioxide via expired air. After oral administration, chondroitin sulphate is absorbed from the gastrointestinal tract and is distributed into the joint space.

DOSAGE AND ADMINISTRATION:

Adult:

One sachet once daily. Dissolve the entire sachet of powder with one (1) cup of room temperature water (250ml) and consume immediately, preferably before meal. For severe osteoarthritis symptoms: Initially treatment of 1 sachet once daily is recommended during a period of at least 8-12 weeks. The treatment of osteoarthritis should be repeated every other 6 months or less (according to medical prescription). Axcel Glucosatin Dry Powder treats the underlying cause of osteoarthritis and the therapeutic effect can only be seen after 2-3 weeks. Therefore, it is advisable to take an analgesic or anti-inflammatory drug if required during the first 2-3 weeks if therapy.

Children:

Safety and effectiveness have not been established in children.

CONTRAINDICATIONS:

Contraindicated in patients sensitive to the active ingredients or any of the excipients. As the active ingredient is obtained from seafood (shellfish), the product should not be given to patients who are allergic to shellfish.

PRECAUTIONS:

Glucosamine treats the underlying cause of osteoarthritis and the therapeutic effect can only be seen after 2-3 weeks. Therefore, it is advisable to take an analgesic/anti-inflammatory drug if required during the first 2-3 weeks of therapy with glucosamine. Administration during the first three months of pregnancy must be avoided. Safety and effectiveness have not been established in children therefore children should avoid using glucosamine. The administration in patients with severe hepatic or renal insufficiency should be made under medical supervision. Derived from seafood, therefore should not be given to patients who are allergic to shellfish. A doctor should be consulted in order to exclude the presence of other joint conditions/diseases for which an alternative treatment should be considered. Glucosamine may affect insulin sensitivity and glucose tolerance. Patients with type 2 diabetes and patients who are overweight or having glucose tolerance problems should monitor their blood sugar levels regularly when taking any products containing glucosamine. Unsuitable for phenylketonurics.

Effects on Ability to Drive and Use Machines:

No effects on the ability to drive or to operate machines are expected.

Use in Pregnancy and Lactation:

Available evidence is inconclusive or inadequate for use in pregnant or lactating mothers. Until more information is available, this product should only be used under medical supervision in pregnancy and lactating mother if the potential benefit to the mother justifies the potential risk to the fetus. Administration during

the first 3 months of pregnancy must be avoided.

SIDE EFFECTS:

Cardiovascular: Peripheral oedema, tachycardia were reported in a few patients following larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.

Central nervous system: Drowsiness, headache, insomnia have been observed rarely during therapy (less than 1%).

Gastrointestinal: Nausea, vomiting, diarrhoea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described rarely during therapy with glucosamine.

Skin: Skin reactions such as erythema and pruritus have been reported with therapeutic administration of glucosamine.

DRUG INTERACTION:

Effects on glucose metabolism & antidiabetic agents:

It has been hypothesized that glucosamine may impair insulin secretion through competitive inhibition of glucokinase in pancreatic beta cells and/or alteration of peripheral glucose uptake. Glucosamine may increase insulin resistance and consequently affect glucose tolerance. It may reduce antidiabetic agent effectiveness e.g. when used with these antidiabetic agent : Acarbose, Acetohexamide, Chlorpropamide, Glipizide, Glyburide, Metformin, Miglitol, Pioglitazone, Repaglinide, Rosiglitazone, Glimepiride, Tolbutamide and Troglitazone. Glucosamine is likely safe in patients with well-controlled diabetes (HbA1c less than 6.5%) taking one or two oral antidiabetic medications or controlled by diet only. In patients with higher HbA1c levels or those taking insulin, monitor blood glucose levels closely/more frequently. Reduced effectiveness when used with glucosamine: Doxorubicin, Etoposide and Teniposide.

Warfarin:

Elevation of International Normalized Ratio serum values and potentiation of anticoagulant effects. If concomitant therapy is necessary, the patient's INR should be more closely monitored.

OVERDOSAGE AND TREATMENT:

No cases of accidental or intentional overdose are known. The animal acute and chronic toxicological studies indicate that toxic effects and symptoms of toxicity are not likely to occur, even after high overdoses.

STORAGE:

Store in a dry place, below 30°C. Protect from moisture.

KEEP OUT OF REACH OF CHILDREN JAUHI DARI KANAK-KANAK

PACK QUANTITIES:

Available in 5g x 30 sachets per box.

Further information can be obtained from pharmacist, physician or the manufacturer.

Manufactured By and Product Registration Holder



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MAL18016082X

Regn - 021121
Date of revision: 02/11/21